MIL-E-1/436B
21 NOV 1975
SUPERSEDING
MIL-E-1/436A
28 December 1967

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 12BA6

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Pentode, miniature, RF remote cutoff

Outline --- 5-2 (EIA)

Base --- E7-1

Envelope --- T5-1/2

Cathode --- Coated unipotential

Base connections:

Pin No.	---	1	2	3	4	5	6	7
Element	---	g1	g3 (int sd)	h	h	p	g2	k

ABSOLUTE RATINGS:

Parameter:	Ef	Eb	Ec1	Ec2	Ec3	Ehk	Rk	Pp	Pg2	Alt
Unit:	V	Vdc	Vdc	Vdc	Vdc	v	ohms	W	W	ft
Maximum:	13.9	330	---	135	---	100	--	3.3	0.7	(See note 2)
Minimum:	11.3	---	---	---	---	---	--	---	---	
TEST CONDITIONS:	12.6	250	---	100	0	(See note 1)	68	---	---	---

GENERAL:

Qualification - Required

12BA6

FSC 5960

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS		UNIT
						Min	Max	
	Quality conformance inspection, part 1							
1266	Total grid current	Ec1 = -1.5 Vdc See note 3	0.65	II	Ic1	0	-1.0	μAdc
1256	Electrode current (anode)		0.65	II	Ib	8.5	13.5	mAdc
1306	Transconductance (1)	Ck = 1,000 f	0.65	II	Sm	3,600	5,200	μmhos
1231	Emission	Eb = Ec1 = Ec2 = 20 Vdc; Rk = 0 See note 3	0.65	II	Is	60	---	mAdc
1201	Short and discontinuity detection		0.4	II	---	---	---	---
Appendix D, 30(b)	Dimensions	Outline 5-2; dimensions A, C, D and E	1.0	I	---	---	---	---
	Quality conformance inspection, part 2							
1211	Insulation of electrodes		4.0	S3	R	10	---	Meg
1256	Electrode current (screen)		6.5	S3	Ic2	2.8	5.6	mAdc
1031	Low-frequency vibration	Rp = 2,000 ohms; Ck = 1,000 μf	6.5	S3	Ep	---	400	mVac
1301	Heater current		6.5	S3	If	138	162	mA
1336	Heater-cathode leakage		6.5	S3	Ihk	---	20	μAdc
1306	Transconductance (2)	Ec1 = -20 Vdc	6.5	S3	Sm	5	100	μmhos
1331	Direct-interelectrode capacitance	Shield No. 316	6.5	S3	Cgp Cin Cout	--- 4.4 3.5	0.0035 6.6 6.5	pf pf pf
1121	Base strain		---	---	---	---	---	---
2126	Glass strain		4.0	I	---	---	---	---
1105	Permanence of marking		---	---	---	---	---	---
	Quality conformance inspection, part 3							
1501	Intermittent life-test provisions	Group A; Ehk = 100 V; Rg1 = 0.05 Meg(min)	---	---	---	---	---	---
	Intermittent life-test end point: (500 hours)							
1306	Tranconductance (1)		---	---	Sm	2,800	---	μmhos

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NOTES:

1. The reference point for heater-cathode potential shall be the positive terminal of the cathode resistor, unless otherwise specified.
2. See in the basic document "Reduced pressure (altitude) rating", and altitude, maximum peak voltage.
3. This test to be performed at the conclusion of the holding period.
4. The tube described by this TSS is interchangeable with the tube procured to the previous issue.

Custodians:

Army - EL
Navy - EC
Air Force - 85

Preparing activity:

Air Force - 85

Review activities:

Army - EL
Navy - EC
Air Force - 80
DSA - ES

(Project No. 5960-2830)

User activities:

Army - MU, ME
Navy - AS, OS, MC, CG
Air Force - 11

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OMB Approval
No. 22-R25

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DOCUMENT IDENTIFIER AND TITLE
MIL-E-1/436B Electron Tube, Receiving Type 12BA6 FSC 5960

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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